

Hilton Stammbach 1

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1 Modules

1. Suppose $\omega : \Lambda \rightarrow \text{Hom}(A, A)$ is a ring homomorphism. Denote by $\lambda x = (\omega(\lambda))(x)$ for all λ, x . Then the following axioms hold:

M1. $(\lambda_1 + \lambda_2)x = \lambda_1 x + \lambda_2 x$.

M2. $1x = x$.

M3. $(\lambda_1 \lambda_2)x = \lambda_1(\lambda_2 x)$.

M4. $\lambda(x_1 + x_2) = \lambda x_1 + \lambda x_2$.

Proof:

$$\begin{aligned} (\lambda_1 + \lambda_2)x &= \omega(\lambda_1 + \lambda_2)(x) = (\omega(\lambda_1) + \omega(\lambda_2))(x) \\ &= \omega(\lambda_1)x + \omega(\lambda_2)x = \lambda_1 x + \lambda_2 x \end{aligned}$$

$$1x = \omega(1)x = 1_A x = x$$

$$(\lambda_1 \lambda_2)x = \omega(\lambda_1 \lambda_2)(x) = \omega(\lambda_1)(\omega(\lambda_2)(x)) = \lambda_1(\lambda_2 x)$$

$$\begin{aligned} \lambda(x_1 + x_2) &= \omega(\lambda)(x_1 + x_2) = \omega(\lambda)x_1 + \omega(\lambda)x_2 \\ &= \lambda x_1 + \lambda x_2 \end{aligned}$$

Conversely, suppose M1 to M4 are satisfied. Define $\omega : \Lambda \rightarrow \text{Hom}(A, A)$ by $\omega(\lambda)(x) = \lambda x$. $\omega(\lambda)$ is a homomorphism:

$$\omega(\lambda)(x_1 + x_2) = \lambda(x_1 + x_2) = \lambda x_1 + \lambda x_2 = \omega(\lambda)x_1 + \omega(\lambda)x_2.$$

$\omega(1)$ is sent to the identity of $\text{Hom}(A, A)$:

$$\omega(1)(x) = 1x = x = 1_A x.$$

ω preserves addition:

$$\omega(\lambda_1 + \lambda_2)(x) = (\lambda_1 + \lambda_2)x = \lambda_1 x + \lambda_2 x = \omega(\lambda_1)x + \omega(\lambda_2)x.$$

ω preserves multiplication:

$$\omega(\lambda_1 \lambda_2)x = (\lambda_1 \lambda_2)x = \lambda_1(\lambda_2 x) = \omega(\lambda_1)\omega(\lambda_2)x.$$

The opposite ring is given by $\lambda_1^{opp} \lambda_2^{opp} = (\lambda_2 \lambda_1)^{opp}$. We have

$$\begin{aligned} (\lambda_1^{opp} \lambda_2^{opp}) \lambda_3^{opp} &= (\lambda_2 \lambda_1)^{opp} \lambda_3^{opp} = (\lambda_3 (\lambda_2 \lambda_1))^{opp} \\ &= ((\lambda_3 \lambda_2) \lambda_1)^{opp} = \lambda_1^{opp} (\lambda_3 \lambda_2)^{opp} = \lambda_1^{opp} (\lambda_2^{opp} \lambda_3^{opp}), \end{aligned}$$

so multiplication is associative.

Claim: A right module is a homomorphism $\omega : \Lambda^{opp} \rightarrow \text{Hom}(A, A)$. Denote by $x\lambda = \omega(\lambda^{opp})(x)$. The axioms are:

M1'. $(x_1 + x_2)\lambda = x_1\lambda + x_2\lambda$.

M2'. $x(\lambda_1 + \lambda_2) = x\lambda_1 + x\lambda_2$.

M3'. $x(\lambda_1 \lambda_2) = (x\lambda_1)\lambda_2$.

M4'. $x1 = x$.

We show one direction. We have

$$\begin{aligned} (x_1 + x_2)\lambda &= \omega(\lambda^{opp})(x_1 + x_2) = \omega(\lambda^{opp})(x_1) + \omega(\lambda^{opp})(x_2) = x_1\lambda + x_2\lambda \\ x(\lambda_1 + \lambda_2) &= \omega((\lambda_1 + \lambda_2)^{opp})x = \omega(\lambda_1^{opp})x + \omega(\lambda_2^{opp})x = x\lambda_1 + x\lambda_2 \\ x(\lambda_1 \lambda_2) &= \omega((\lambda_1 \lambda_2)^{opp})(x) = \omega(\lambda_2^{opp} \lambda_1^{opp})(x) = \omega(\lambda_2^{opp})\omega(\lambda_1^{opp})(x) \\ &= \omega(\lambda_2^{opp})(x\lambda_1) = (x\lambda_1)\lambda_2 \\ x1 &= \omega(1^{opp})x = 1_A x = x \end{aligned}$$

Suppose we have the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A' & \xrightarrow{\mu} & A & \xrightarrow{\varepsilon} & A'' \longrightarrow 0 \\ & & \downarrow \alpha' & & \downarrow \alpha & & \downarrow \alpha'' \\ 0 & \longrightarrow & B' & \xrightarrow{\mu'} & B & \xrightarrow{\varepsilon'} & B'' \longrightarrow 0 \end{array}$$

such that the rows are exact. If two of $\alpha', \alpha, \alpha''$ are isomorphisms, so is the third.

Proof: Suppose α, α'' are isomorphisms. Let $b' \in B'$. Let $b = \mu' b'$ and pick $a \in A$ such that $\alpha a = b$. Note that $\varepsilon a = \varepsilon \alpha^{-1} b = (\alpha'')^{-1} \varepsilon' b = (\alpha'')^{-1} \varepsilon' \mu' b' = 0$, so write $a = \mu a'$ for some $a' \in A'$. Then $\mu' \alpha' a' = \alpha \mu a' = \alpha a = b = \mu' b'$, so $\mu'(\alpha' a' - b') = 0 \rightarrow b' = \alpha' a'$. Suppose $\alpha' a' = 0$. Then $0 = \mu' \alpha' a' = \alpha \mu a'$. Since α, μ are mono, $a' = 0$.

Now suppose α', α'' are isoms. Let $b \in B$. Since ε, α'' are epi, pick $a \in A$ such that $b'' = \varepsilon' b = \alpha'' \varepsilon a$. Then $\varepsilon' \alpha a = \alpha'' \varepsilon a = \varepsilon' b$, so $\varepsilon'(\alpha a - b) = 0 \rightarrow \alpha a - b = \mu' b'$ for some $b' \in B'$. Write $b' = \alpha' a'$. Then $b = \alpha a + \mu' b' = \alpha a + \mu' \alpha' a' = \alpha a + \alpha \mu a' = \alpha(a + \mu a')$, so α is epi. Suppose $\alpha a = 0$. Then $0 = \varepsilon' \alpha a = \alpha'' \varepsilon a \rightarrow \varepsilon a = 0$. Write $a = \mu a'$. Then $0 = \alpha \mu a' = \mu' \alpha' a'$. Since μ', α' are mono, we have $a' = 0$, so $a = \mu a' = 0$, so α is mono.

Suppose α', α are isoms. Let $b'' \in B''$. Since α, ε' are epi, pick $a \in A$ such

that $\varepsilon'\alpha a = b''$. Let $a'' = \varepsilon a$. Then $\alpha''a'' = \alpha''\varepsilon a = \varepsilon'\alpha a = b''$, so α'' is epi. Suppose $\alpha''a'' = 0$. Let $a'' = \varepsilon a$. Then $\varepsilon'\alpha a = 0$ implies $\alpha a = \mu'b'$ for some $b' \in B'$. Pick $a' \in A'$ such that $\alpha'a' = b'$. Then $\alpha a = \mu'\alpha'a' = \alpha\mu a' \rightarrow a = \mu a'$, so $a'' = \varepsilon a = \varepsilon\mu a' = 0$, so α'' is mono.

- (i). Take $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ and $0 \rightarrow \mathbb{Z} \xrightarrow{1} \mathbb{Z} \rightarrow 0 \rightarrow 0$.
- (ii). Take $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \oplus (\mathbb{Z}/2\mathbb{Z}) \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ and $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$.
- (iii). Take $0 \rightarrow 2\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ and $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$.

Suppose $mA = 0$. From the usual group structure, we have a ring homomorphism $\omega : \mathbb{Z} \rightarrow \text{End}(A, A)$. If $n = km$ for some $k \in \mathbb{Z}$, then $(km)a = k(ma) = k0 = 0$, so $m\mathbb{Z} \leq \ker \omega$. ω factors through $\mathbb{Z}/m\mathbb{Z}$ to $\omega' : \mathbb{Z}/m\mathbb{Z} \rightarrow \text{End}(A, A)$, so A is a $\mathbb{Z}/m\mathbb{Z}$ -module. Conversely, if $\omega' : \mathbb{Z}/m\mathbb{Z} \rightarrow \text{End}(A, A)$ is a ring homomorphism, then since $0 = m$ in $\mathbb{Z}/m\mathbb{Z}$, we have that $mA = 0$.

Let $0 \rightarrow A' \xrightarrow{\mu} A \xrightarrow{\varepsilon} A'' \rightarrow 0$ be a short exact sequence of abelian groups. Show that the following are equivalent:

- (i). The sequence is pure.
- (ii). The induced sequence $0 \rightarrow A'_n \xrightarrow{\mu_n} A_n \xrightarrow{\varepsilon_n} A''_n \rightarrow 0$ is exact for all n .
- (iii). Given $a'' \in A''$ such that $na'' = 0$, there exists $a \in A$ such that $\varepsilon a = a''$ and $na = 0$.

Proof:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A' & \xrightarrow{\mu} & A & \xrightarrow{\varepsilon} & A'' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A'_n & \xrightarrow{\mu_n} & A_n & \xrightarrow{\varepsilon_n} & A''_n & \longrightarrow & 0 \end{array}$$

(i) implies (iii). Suppose $a'' \in A''$ such that $na'' = 0$. Pick $a \in A$ such that $\varepsilon a = a''$. Then $\varepsilon na = n\varepsilon a = na'' = 0$, so $na = \mu a'$ for some $a' \in A'$. Since $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ is pure, pick $a'_1 \in A'$ such that $a' = na'_1$. Then $\mu a' = \mu na'_1 = n\mu a'_1$, so $n(a - \mu a'_1) = 0$. Note that $\varepsilon(a - \mu a'_1) = \varepsilon a = a''$, so we're done.

(iii) implies (i). Let $a' \in A'$ such that $\mu a' = na$ for some $a \in A$. Then $0 = \varepsilon \mu a' = \varepsilon na = n\varepsilon a$. Pick $a_1 \in A$ such that $\varepsilon a_1 = \varepsilon a$ and $na_1 = 0$. Then $\varepsilon(a - a_1) = 0$, so $a - a_1 = \mu a'_1$ for some $a'_1 \in A'$. Multiplying by n , we have $\mu a' = na = n\mu a'_1$. Since μ is mono, we have $a' = na'_1$.

(i), (iii) imply (ii). Since $\varepsilon\mu = 0$ and the projection $A' \rightarrow A'_n$ is surjective, we have that $\varepsilon_n\mu_n = 0$. Suppose $\varepsilon_n\bar{a} = 0$, so $\varepsilon a = na''$ for some $a'' \in A''$. Pick $a_1 \in A$ such that $\varepsilon a_1 = a''$. Then $\varepsilon(a - na_1) = 0$. Pick $a' \in A'$ such that $\mu a' = a - na_1$, so $\mu_n\bar{a}' = \bar{a}$, so the sequence is exact at A .

We show that μ_n is mono and ε_n is epi. Suppose $\mu_n\bar{a}' = 0$, so $\mu a' = na$

for some $a \in A$. By (i), there exists $a'_1 \in A'$ such that $a' = na'_1$, so $\bar{a} = 0$ in A_n . Since the projection $A'' \rightarrow A''_n$ and ε are epi, so is ε_n .

(ii) implies (i). Suppose $a' \in A'$ such that $\mu a' = na$ for some $a \in A$. Then $\mu_n \bar{a}' = 0$. Since μ_n is mono, we have $\bar{a}' = 0$, ie $a' \in nA'$.

Let $\beta : B_1 \rightarrow B_2$ be a hom of Λ -modules. For all Λ -modules A , and $\varphi : A \rightarrow B_1$, we have the assignment $\varphi \mapsto \beta\varphi$, so a map $\beta_* = \text{Hom}_\Lambda(A, \beta) : \text{Hom}_\Lambda(A, B_1) \rightarrow \text{Hom}_\Lambda(A, B_2)$. I claim that β_* is a group homomorphism. Indeed,

$$\beta_*(\varphi_1 + \varphi_2) = \beta(\varphi_1 + \varphi_2) = \beta\varphi_1 + \beta\varphi_2 = \beta_*(\varphi_1) + \beta_*(\varphi_2).$$

If $\beta : B_1 \rightarrow B_2, \beta' : B_2 \rightarrow B_3$, we have $(\beta'\beta)_* = \beta'_*\beta_*$. Indeed,

$$(\beta'\beta)_*(\varphi) = \beta'\beta\varphi = \beta'_*(\beta\varphi) = \beta'_*\beta_*(\varphi).$$

If $\beta : B \rightarrow B$ is the identity, then $\beta_* : \text{Hom}(A, B) \rightarrow \text{Hom}(A, B)$ is also the identity. Indeed, $\beta_*(\varphi) = \beta\varphi = \varphi = 1\varphi$.

Suppose $\alpha : A_2 \rightarrow A_1$ is a homomorphism and B is fixed. Let $\varphi : A_1 \rightarrow B$ be a module homomorphism. Define $\alpha_*(\varphi) = \varphi\alpha$, so $\alpha_* = \text{Hom}_\Lambda(\alpha, B) : \text{Hom}_\Lambda(A_1, B) \rightarrow \text{Hom}_\Lambda(A_2, B)$. Note that $\alpha_*(\varphi_1 + \varphi_2) = (\varphi_1 + \varphi_2)\alpha = \varphi_1\alpha + \varphi_2\alpha = \alpha_*(\varphi_1) + \alpha_*(\varphi_2)$, so α_* is a group homomorphism.

(i). If $\alpha : A_2 \rightarrow A_1$ and $\alpha' : A_3 \rightarrow A_2$, then $(\alpha'\alpha)_* = \alpha_*\alpha'_*$. Indeed, $(\alpha'\alpha)_*(\varphi) = \varphi\alpha'\alpha = \alpha_*(\varphi\alpha') = \alpha_*\alpha'_*(\varphi)$.

(ii). If $\alpha : A \rightarrow A$ is the identity, then $\alpha_*(\varphi) = \varphi\alpha = \varphi$, so α_* is the identity.

$\text{Hom}_\Lambda(-, B)$ is a contravariant functor from $\Lambda\text{-mod}$ to the category of abelian groups.

Let A, B', B, B'' be Λ -modules such that $0 \rightarrow B' \xrightarrow{\mu} B \xrightarrow{\varepsilon} B'' \rightarrow 0$ is exact. Then $0 \rightarrow \text{Hom}_\Lambda(A, B') \xrightarrow{\mu_*} \text{Hom}_\Lambda(A, B) \xrightarrow{\varepsilon_*} \text{Hom}_\Lambda(A, B'')$ is exact.

Proof: I claim that μ_* is mono. Suppose $\varphi : A \rightarrow B'$ is a Λ -hom such that $\mu_*\varphi = \mu\varphi = 0$. Since μ is mono, we have $\varphi = 0$. Now we show exactness at $\text{Hom}_\Lambda(A, B)$. Suppose $\varphi : A \rightarrow B$ such that $\varepsilon_*\varphi = \varepsilon\varphi = 0$. Then $\text{im } \varphi \leq \ker \varepsilon = \text{im } \mu$. The restriction $\mu : B' \rightarrow \text{im } \mu$ is a bimorphism and thus an isom (in the category of $\Lambda\text{-mod}$). Thus, we have a map $\psi : A \rightarrow B'$ such that the following diagrams commute

$$\begin{array}{ccc} & A & \\ \swarrow \psi & \downarrow \varphi & \\ B' & \xrightarrow{\mu} & \text{im } \mu \end{array} \quad \begin{array}{ccc} & A & \\ \swarrow \psi & \downarrow \varphi & \\ B' & \xrightarrow{\mu} & B \end{array}$$

So $\varphi = \mu\psi = \mu_*\psi$.

Suppose $0 \rightarrow A' \xrightarrow{\mu} A \xrightarrow{\varepsilon} A'' \rightarrow 0$ is exact. Then $0 \rightarrow \text{Hom}_\Lambda(A'', B) \xrightarrow{\varepsilon_*} \text{Hom}_\Lambda(A', B) \xrightarrow{\mu_*} \text{Hom}_\Lambda(A, B)$ is exact.

Proof: Suppose $\varphi : A'' \rightarrow B$ such that $\varepsilon_*\varphi = \varphi\varepsilon = 0$. Since ε is epi, we have $\varphi = 0$, so ε_* is mono. We have $\mu_*\varepsilon_* = (\varepsilon\mu)_* = 0_x = 0$. Suppose $\mu_*\varphi = 0$. Then $\varphi\mu = 0$, so $\ker \varepsilon = \text{im } \mu \leq \ker \varphi$. Let $p : A \rightarrow A/\ker \varepsilon$ denote the projection and $\bar{\varepsilon} : A/\ker \varepsilon \rightarrow A''$ the isomorphism, so $\varepsilon = \bar{\varepsilon}p$. Since φ vanishes on $\ker \varepsilon$, we descend to the map $\bar{\varphi} : A/\ker \varepsilon \rightarrow B$ such that $\bar{\varphi}p = \varphi$. Let $\psi = \bar{\varphi}(\bar{\varepsilon})^{-1} : A'' \rightarrow B$. Note that $\psi\varepsilon = \psi\bar{\varepsilon}p = \bar{\varphi}p = \varphi$, so $\varepsilon_*(\psi) = \varphi$.

If $0 \rightarrow B' \xrightarrow{\mu} B \xrightarrow{\varepsilon} B'' \rightarrow 0$ is exact, then ε_* need not be surjective. Take $\Lambda = \mathbb{Z}, B' = B = \mathbb{Z}, B'' = \mathbb{Z}/n\mathbb{Z}, A = \mathbb{Z}/n\mathbb{Z}$ and $\mu = \times n$. Then we have $0 \rightarrow 0 \xrightarrow{\mu_*} 0 \xrightarrow{\varepsilon_*} \mathbb{Z}/n\mathbb{Z} \rightarrow 0$, so ε_* is not surjective.

If $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ is exact, then μ_* need not be surjective. Take $\Lambda = \mathbb{Z}, A' = A = \mathbb{Z}, A'' = \mathbb{Z}/n\mathbb{Z}, B = \mathbb{Z}/n\mathbb{Z}$ and $\mu = \times n$. Then we have $0 \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$. Suppose $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$. Then $\mu_*\varphi = \varphi n = n\varphi = 0$, so μ_* need not be surjective.

Let A, B be Λ modules for Λ commutative. Define the action of Λ on $\text{Hom}_\Lambda(A, B)$ by $(\lambda\varphi)(a) = \varphi(\lambda a)$. Then this action is well-defined and gives $\text{Hom}_\Lambda(A, B)$ a Λ module structure. Show for noncommutative Λ that this definition doesn't work.

M1: $\lambda(\varphi_1 + \varphi_2)(a) = (\varphi_1 + \varphi_2)(\lambda a) = \varphi_1(\lambda a) + \varphi_2(\lambda a) = (\lambda\varphi_1)(a) + (\lambda\varphi_2)(a)$.
M2: $(\lambda_1 + \lambda_2)\varphi(a) = \varphi((\lambda_1 + \lambda_2)a) = \varphi(\lambda_1 a + \lambda_2 a) = \varphi(\lambda_1 a) + \varphi(\lambda_2 a) = (\lambda_1\varphi)(a) + (\lambda_2\varphi)(a)$. M3: $((\lambda_1\lambda_2)\varphi)(a) = \varphi((\lambda_1\lambda_2)a) = \varphi((\lambda_2\lambda_1)a) = \varphi(\lambda_2(\lambda_1 a)) = (\lambda_1(\varphi\lambda_2))(a) = (\lambda_1(\lambda_2\varphi))(a)$. M4: $(1\varphi)(a) = \varphi(1a) = \varphi a$.

Suppose Λ is not commutative. Take $A = B = \Lambda$ and $a = 1$. Pick $\lambda_1, \lambda_2 \in \Lambda$ such that $\lambda_1\lambda_2 \neq \lambda_2\lambda_1$. Then $(\lambda_1\lambda_2)\varphi(a) = \varphi((\lambda_1\lambda_2)a) = (\lambda_1(\lambda_2\varphi))(a) = (\lambda_2\varphi)(\lambda_1 a) = \varphi(\lambda_2(\lambda_1 a))$. Take $\varphi = 1_\Lambda$ and $a = 1$. Then we are reduced to $\lambda_1\lambda_2 = \lambda_2\lambda_1$, absurd.

$\text{Hom}(\mathbb{Z}, \mathbb{Z}_n) = \mathbb{Z}_n$. $\text{Hom}(\mathbb{Z}_m, \mathbb{Z}_n) = \mathbb{Z}_{\gcd(n, m)}$. Let $\varphi \in \text{Hom}(\mathbb{Z}_m, \mathbb{Z}_n)$ and $\bar{y} = \varphi(\bar{1})$, so $m\bar{y} = 0 \rightarrow n \mid my \rightarrow n/\gcd(n, m) \mid y$. Thus, we want $((n/\gcd(n, m))\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_{\gcd(n, m)})$. $\text{Hom}(\mathbb{Z}_m, \mathbb{Z}) = 0$. Suppose $\varphi : \mathbb{Q} \rightarrow \mathbb{Z}$. Let $q \neq 0$ and suppose $\varphi q = n \neq 0$. Then $2n\varphi(q/(2n)) = \varphi(q) = n \rightarrow 2\varphi(q/(2n)) = 1$, absurd. Thus, $\text{Hom}(\mathbb{Q}, \mathbb{Z}) = 0$. $\text{Hom}(\mathbb{Q}, \mathbb{Q}) = \mathbb{Q}$. Suppose $\varphi 1 = q$. Then $m\varphi(1/m) = \varphi 1 = q \rightarrow \varphi(1/m) = q/m$. Then $\varphi(n/m) = n\varphi(1/m) = qn/m$.

$0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ is pure iff $\text{Hom}(\mathbb{Z}_n, -)$ preserves exactness for all n . I claim that the sequence is pure iff $0 \rightarrow T_m(A') \xrightarrow{T_m(\mu)} T_m(A) \xrightarrow{T_m(\varepsilon)} T_m(A'') \rightarrow 0$ is exact for all m . Suppose the sequence is pure. Note that the maps $T_m(\mu), T_m(\varepsilon)$ are well-defined. Since μ is injective, so is $T_m(\mu)$. Suppose $a'' \in A''$ such that $ma'' = 0$. By definition, there exists $a \in A$ such that $\varepsilon a = a''$ and

$ma = 0$, ie $T_m(\varepsilon)a = a''$, so $T_m(\varepsilon)$ is surjective. Suppose $a \in A$ such that $ma = 0$ and $\varepsilon a = 0$. By exactness, let $a' \in A'$ such that $\mu a' = a$, so $0 = ma = m\mu a' = \mu ma' \rightarrow ma' = 0$, so $a' \in T_m(A)$, so the sequence in T_m is exact at $T_m(A)$ (we have $T_m(\varepsilon)T_m(\mu) = T_m(\varepsilon\mu) = T_m(0) = 0$).

Now suppose the sequence in T_m is exact. Let $a'' \in A''$ such that $ma'' = 0$. Since $T_m(\varepsilon)$ is surjective, there exists $a \in A$ such that $ma = 0$ and $\varepsilon a = a''$, so the sequence is pure.

$0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ is (exact) pure iff $0 \rightarrow \text{Hom}(\mathbb{Z}_n, A') \rightarrow \text{Hom}(\mathbb{Z}_n, A) \rightarrow \text{Hom}(\mathbb{Z}_n, A'') \rightarrow 0$ is exact.

Proof: Since the sequence is exact, we have that $0 \rightarrow \text{Hom}(\mathbb{Z}_n, A') \rightarrow \text{Hom}(\mathbb{Z}_n, A) \rightarrow \text{Hom}(\mathbb{Z}_n, A'')$ is exact. It suffices to show $\varepsilon_* : \text{Hom}(\mathbb{Z}_n, A) \rightarrow \text{Hom}(\mathbb{Z}_n, A'')$ is surjective. Let $\varphi : \mathbb{Z}_n \rightarrow A''$. Note $n\varphi(\bar{1}) = \varphi 0 = 0$. Pick $a \in A$ such that $\varepsilon a = a''$ and $na = 0$. Define the map $\psi : \mathbb{Z}_n \rightarrow A$ by $\psi(\bar{1}) = a$. Then $\varepsilon_*(\psi)(\bar{1}) = \varepsilon\psi(\bar{1}) = \varepsilon a = a''$, so $\varepsilon_*\psi = \varphi$.

Now suppose ε_* is surjective. Let $a'' \in A''$ such that $na'' = 0$. Define the map $\varphi : \mathbb{Z}_n \rightarrow A''$ by $\varphi(\bar{1}) = a''$. Pick $\psi : \mathbb{Z}_n \rightarrow A$ such that $\varepsilon_*\psi = \varphi$. Take $a = \psi(\bar{1})$, so $\varepsilon a = \varepsilon\psi(\bar{1}) = \varphi(\bar{1}) = a''$ and $na = 0$. Hence, the sequence is pure.

2 Sums and Products

Let A, B, M be Λ modules and $\psi_A : A \rightarrow M, \psi_B : B \rightarrow M$ module homomorphisms. Then there exists a unique map $\psi : A \oplus B \rightarrow M$ such that $\psi i_A = \psi_A$ and $\psi i_B = \psi_B$.

Proof: Define $\psi : A \oplus B \rightarrow M$ by $\psi(a, b) = \psi_A(a) + \psi_B(b)$. ψ is a homomorphism since

$$\begin{aligned} \psi(a, b) + \psi(a', b') &= \psi_A(a) + \psi_B(b) + \psi_A(a') + \psi_B(b') \\ &= \psi_A(a + a') + \psi_B(b + b') = \psi(a + a', b + b') \\ \lambda\psi(a, b) &= \lambda(\psi_A(a) + \psi_B(b)) = \lambda\psi_A(a) + \lambda\psi_B(b) \\ &= \psi_A(\lambda a) + \psi_B(\lambda b) = \psi(\lambda a, \lambda b). \end{aligned}$$

The map satisfies $\psi_A = \psi i_A$ and $\psi_B = \psi i_B$. Suppose ψ' is another such map. Then

$$\begin{aligned} \psi'(a, b) &= \psi' i_A(a) + \psi' i_B(b) = \psi_A(a) + \psi_B(b) \\ &= \psi i_A(a) + \psi i_B(b) = \psi(a, b). \end{aligned}$$

Let $\{\psi_j : A_j \rightarrow M\}$ be homomorphisms. Then there exists a unique homomorphism $\psi : \bigoplus_{j \in J} A_j \rightarrow M$ such that $\psi_j = \psi i_j$.

Proof: Define $\psi((a_j)_{j \in J}) = \sum_{j \in J} \psi_j(a_j)$. Note that the sum is well-defined since all but finitely many of the terms are zero.

Let $\{A_j\}$ be a collection of modules such that there exist $\{\iota_j : A_j \rightarrow A\}$ so that for all $\{\psi_j : A_j \rightarrow M\}$, there exists $\psi : A \rightarrow M$ such that $\psi\iota_j = \psi_j$.

Suppose we have the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A' & \xrightarrow{\mu} & A & \xrightarrow{\varepsilon} & A'' \longrightarrow 0 \\ & & \downarrow \alpha' & & \downarrow \alpha & & \downarrow \alpha'' \\ 0 & \longrightarrow & B' & \xrightarrow{\mu'} & B & \xrightarrow{\varepsilon'} & B'' \longrightarrow 0 \end{array}$$

Then we have the exact sequence $0 \rightarrow A \xrightarrow{\{\varepsilon, \alpha\}} A'' \oplus B \xrightarrow{\langle \alpha'', -\varepsilon' \rangle} B'' \rightarrow 0$.

Proof: (1). $\{\varepsilon, \alpha\}$ is injective. Suppose $\alpha a = 0$ and $\varepsilon a = 0$. Let $a = \mu a'$, so $0 = \alpha a = \alpha \mu a' = \mu' \alpha' a'$. Since μ', α' are injective, we have $a' = 0$, so $a = \mu a = 0$.

(2). $\langle \alpha'', -\varepsilon' \rangle$ is surjective. Let $b'' \in B''$. Pick b such that $\varepsilon' b = b''$, so $\langle \alpha'', -\varepsilon' \rangle(0, -b) = \varepsilon' b = b''$.

(3). $\ker(\langle \alpha'', -\varepsilon' \rangle) = \text{im } \{\varepsilon, \alpha\}$. Note that $\langle \alpha'', -\varepsilon' \rangle \{\varepsilon, \alpha\}(a) = \langle \alpha'', -\varepsilon' \rangle(\varepsilon a, \alpha a) = \alpha'' \varepsilon a - \varepsilon' \alpha a = 0$. Suppose $\langle \alpha'', -\varepsilon' \rangle(a'', b) = 0$. Then $\alpha'' a'' = \varepsilon' b$. Pick a such that $\varepsilon a = a''$. Then $\varepsilon' b = \alpha'' \varepsilon a = \varepsilon' \alpha a$, so $\varepsilon'(b - \alpha a) = 0$, so $b - \alpha a = \mu' b'$ for some $b' \in B'$. Let $b' = \alpha' a'$, so $b = \alpha a + \mu' \alpha' a' = \alpha a + \alpha \mu a' = \alpha(a + \mu a')$. Note $\varepsilon(a + \mu a') = \varepsilon a = a''$.

Conversely, I claim that if the above sequence is exact, then α' is an isomorphism. First, we show α' is injective. Suppose $\alpha' a' = 0$. Then $\{\varepsilon, \alpha\} \mu(a') = (\varepsilon \mu a', \alpha \mu a') = (0, \mu' \alpha' a') = (0, 0)$, so $\mu a' = 0 \rightarrow a' = 0$. Now suppose $b' \in B'$. Note that $\langle \alpha'', -\varepsilon' \rangle(0, \mu' b') = -\varepsilon' \mu' b' = 0$. Pick a such that $\varepsilon a = 0$ and $\alpha a = \mu' b'$. Let $a = \mu a'$. Then $\mu' b' = \alpha \mu a' = \mu' \alpha' a' \rightarrow b' = \alpha' a'$.

Use the universal property of the direct sum to show that $(A_1 \oplus A_2) \oplus A_3 \cong A_1 \oplus (A_2 \oplus A_3)$. Let $i_1 : A_1 \rightarrow A_1 \oplus A_2, i_2 : A_2 \rightarrow A_1 \oplus A_2, j_1 : A_2 \rightarrow A_2 \oplus A_3, j_2 : A_3 \rightarrow A_2 \oplus A_3, k_1 : A_1 \oplus A_2 \rightarrow (A_1 \oplus A_2) \oplus A_3, k_2 : A_3 \rightarrow (A_1 \oplus A_2) \oplus A_3$, and $l_1 : A_1 \rightarrow A_1 \oplus (A_2 \oplus A_3), l_2 : A_2 \oplus A_3 \rightarrow A_1 \oplus (A_2 \oplus A_3)$.

Let $\{\varphi_i : A_i \rightarrow M\}$. There exists a unique map $\varphi' : A_1 \oplus A_2 \rightarrow M$ such that $\varphi' i_1 = \varphi_1$ and $\varphi' i_2 = \varphi_2$. There exists a unique map $\varphi : (A_1 \oplus A_2) \oplus A_3 \rightarrow M$ such that $\varphi k_1 = \varphi'$ and $\varphi k_2 = \varphi_3$. Thus, there exists a map φ such that $\varphi k_1 i_1 = \varphi_1, \varphi k_1 i_2 = \varphi_2, \varphi k_2 = \varphi_3$.

Suppose ψ is another map satisfying such properties. We have $\varphi k_1 = \psi k_1$

and $\varphi k_2 = \psi k_2$, so $\varphi = \psi$. Thus, $(A_1 \oplus A_2) \oplus A_3 \cong A_1 \oplus A_2 \oplus A_3$. The other side is similar.

Show that $0 \rightarrow A' \xrightarrow{\alpha'} A \xrightarrow{\alpha''} A'' \rightarrow 0$ splits iff

- (i). $0 \rightarrow \text{Hom}_\Lambda(B, A') \rightarrow \text{Hom}_\Lambda(B, A) \rightarrow \text{Hom}_\Lambda(B, A'') \rightarrow 0$ is exact for all B .
- (ii). $0 \rightarrow \text{Hom}_\Lambda(A'', C) \rightarrow \text{Hom}_\Lambda(A, C) \rightarrow \text{Hom}_\Lambda(A', C) \rightarrow 0$ is exact for all C .
- (iii). There exists $\mu : A'' \rightarrow A$ such that $\langle \alpha', \mu \rangle : A' \oplus A'' \rightarrow A$ is an isomorphism.

Splitting is equivalent to (i). Since the sequence splits, let $k : A'' \rightarrow A$ such that $\alpha'' k = 1_{A''}$. Let $\varphi : B \rightarrow A''$. Then $\alpha''_*(k\varphi) = \alpha'' k \varphi = 1_{A''} \varphi = \varphi$. Conversely, take $B = A''$ and $1_{A''} \in \text{Hom}_\Lambda(A'', A'')$. Pick $k \in \text{Hom}_\Lambda(A'', A)$ such that $\alpha''_* k = 1_{A''}$.

Splitting is equivalent to (ii). Since the sequence splits, let $j : A \rightarrow A'$ such that $j\alpha' = 1_{A'}$. Suppose $\varphi : A' \rightarrow C$. Then $(\alpha')^*(\varphi j) = \varphi j \alpha' = \varphi$. Conversely, take $C = A'$ and $1_{A'} \in \text{Hom}_\Lambda(A', A')$. Pick $j \in \text{Hom}(A, A')$ such that $(\alpha')^*(j) = 1_{A'}$, ie $j\alpha' = 1_{A'}$.

Splitting is equivalent to (iii). Suppose the sequence splits. Take $k : A'' \rightarrow A$. I claim that $\langle \alpha', k \rangle : A' \oplus A'' \rightarrow A$ is an isomorphism. Suppose $\langle \alpha', k \rangle(a', a'') = 0$. Then $\alpha' a' + k a'' = 0$. Applying α'' gives $\alpha'' k a'' = a'' = 0$. Then $\alpha' a' = 0 \rightarrow a' = 0$ since α' is injective.

Now let $a \in A$. Observe $\alpha''(a - k\alpha'' a) = \alpha'' a - \alpha'' a = 0$, so pick $a' \in A'$ such that $a - k\alpha'' a = \alpha' a'$, ie $a = \langle \alpha', k \rangle(a', \alpha'' a)$. Conversely, suppose there exists $\mu : A'' \rightarrow A$ such that $\langle \alpha', \mu \rangle : A' \oplus A'' \rightarrow A$ is an isomorphism. Note that $\langle \alpha', \mu \rangle i_{A'} = \alpha'$, so $i_{A'} = \langle \alpha', \mu \rangle^{-1} \alpha'$, so $1_{A'} = p_{A'} i_{A'} = p_{A'} \langle \alpha', \mu \rangle^{-1} \alpha'$. Take $j = p_{A'} \langle \alpha', \mu \rangle^{-1}$.

3 Free and Projective Modules

A module P is said to be projective if for all surjective $\varepsilon : B \rightarrow C$ and $\gamma : P \rightarrow C$, there exists a map $\beta : P \rightarrow B$ such that $\gamma = \varepsilon \beta$. I claim that equivalently, for all short exact $0 \rightarrow A \xrightarrow{\mu} B \xrightarrow{\varepsilon} C \rightarrow 0$ we have $0 \rightarrow \text{Hom}_\Lambda(P, A) \xrightarrow{\mu_*} \text{Hom}_\Lambda(P, B) \xrightarrow{\varepsilon_*} \text{Hom}_\Lambda(P, C) \rightarrow 0$ is exact, ie ε_* is surjective.

Let $\gamma : P \rightarrow C$. By assumption, there exists $\beta : P \rightarrow B$ such that $\gamma = \varepsilon \beta = \varepsilon_*(\beta)$, so ε_* is surjective. Conversely, if ε_* is surjective, for all $\gamma : P \rightarrow C$, there exists $\beta : P \rightarrow B$ such that $\gamma = \varepsilon_*(\beta) = \varepsilon \beta$.

Claim: $P = \bigoplus_{j \in J} P_j$ is projective iff each P_j is.

Proof: Suppose P is projective. Let $\varepsilon : B \rightarrow C$ be surjective and $\gamma_j : P_j \rightarrow C$.

Define $\gamma_k = 0$ for $k \neq j$ and by extension let $\gamma : P \rightarrow C$. By assumption, pick $\beta : P \rightarrow B$ such that $\gamma = \varepsilon\beta$, so $\gamma_j = \gamma i_j = \varepsilon\beta i_j$. Thus, we take $\beta_j = \beta i_j$. Conversely, suppose each P_j is surjective. Let $\varepsilon : B \rightarrow C$ be surjective and $\gamma : P \rightarrow C$. Let $\gamma_j = \gamma i_j$. Pick β_j such that $\gamma_j = \varepsilon\beta_j$. Let $\beta : P \rightarrow B$ denote the extension of the $\{\beta_j\}$. I claim that $\gamma = \varepsilon\beta$. Indeed, $\gamma i_j = \gamma_j$ and $\varepsilon\beta i_j = \varepsilon\beta_j = \gamma_j$, so by uniqueness we have $\gamma = \varepsilon\beta$.

Let V be a vector space with basis $\{e_i\}_{i \in \mathbb{Z}^+}$. Define the isomorphism $\varphi : V \rightarrow V \oplus V$ by

$$\varphi \left(\sum_i (\lambda_{2i-1} e_{2i-1} + \lambda_{2i} e_{2i}) \right) = \left(\sum_i \lambda_{2i-1} e_i, \sum_i \lambda_{2i} e_i \right).$$

so $V \cong V \oplus V$. Let $\Lambda = \text{Hom}_K(V, V)$. Then

$$\begin{aligned} \Lambda &= \text{Hom}_K(V, V) \cong \text{Hom}_K(V \oplus V, V) \\ &\cong \text{Hom}_K(V, V) \oplus \text{Hom}_K(V, V) \cong \Lambda \oplus \Lambda \end{aligned}$$

Suppose $n < m$. By induction, $\Lambda \cong \Lambda^{m-n+1}$, so $\Lambda^n = \Lambda \oplus \Lambda^{n-1} = \Lambda^{m-n+1} \oplus \Lambda^{n-1} = \Lambda^m$.

Let P, Q be projective. Then there exists projective R such that $P \oplus R \cong Q \oplus R$ is free.

Proof: Take F_1, F_2 free and P', Q' projective such that $F_1 = P \oplus P'$ and $F_2 = Q \oplus Q'$. Let $R = (F_1 \oplus F_2)^{\mathbb{N}_0}$. Then $R = (F_1 \oplus F_2)^{\oplus \mathbb{N}_0} = P^{\oplus \mathbb{N}_0} \oplus (P')^{\oplus \mathbb{N}_0} \oplus Q^{\oplus \mathbb{N}_0} \oplus (Q')^{\oplus \mathbb{N}_0}$ and $R \cong P \oplus R \cong Q \oplus R$.

Claim: $\mathbb{Q}$ is not a free $\mathbb{Z}$ module.

Proof: Suppose FTSOC $\mathbb{Q}$ has a basis $\{e_\alpha\}$. Since $\mathbb{Q} \neq 0$, the basis is nonempty. Suppose $\mathbb{Q}$ has a basis of one element, so $\mathbb{Q} \cong \mathbb{Z}$. Let $\varphi : \mathbb{Q} \rightarrow \mathbb{Z}$ be a group isomorphism. Pick $q \in \mathbb{Q}$ such that $\varphi q = 1$. Then $1 = \varphi(q/2 + q/2) = 2\varphi(q/2)$, absurd. Thus, the basis has at least two elements, say e_1, e_2 . Write $e_1 = m_1/n_1$ and $e_2 = m_2/n_2$ for $m_1, m_2, n_1, n_2 \neq 0$. Then $m_2 n_1 e_1 - m_1 n_2 e_2 = 0$, absurd.

Claim: Suppose we have the following map of exact sequences:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xrightarrow{\mu} & B & \xrightarrow{\varepsilon} & C \longrightarrow 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & A' & \xrightarrow{\mu'} & B' & \xrightarrow{\varepsilon'} & C' \longrightarrow 0 \end{array}$$

such that γ is an isomorphism. Then $0 \rightarrow A \xrightarrow{\{\alpha, \mu\}} A' \oplus B \xrightarrow{\langle \mu', -\beta \rangle} B' \rightarrow 0$ is exact.

Proof: $\{\alpha, \mu\}$ is injective. Suppose $\alpha a = 0$ and $\mu a = 0$. Since μ is injective, we have $a = 0$. $\langle \mu', -\beta \rangle$ is surjective. Let $b' \in B'$. Pick $b \in B$ such

that $\varepsilon'\beta b = \gamma\varepsilon b = \varepsilon'b'$, so $\beta b - b' = \mu'a'$ for some a' , ie $b' = \mu'(-a') - \beta(-b) = \langle \mu', -\beta \rangle(-a', -b)$. The maps $\{\alpha, \mu\}, \langle \mu', -\beta \rangle$ compose to zero. Suppose $\langle \mu', -\beta \rangle(a', b) = 0$, so $\mu'a' = \beta b$. Then $\varepsilon'\beta b = \gamma\varepsilon b = 0 \rightarrow \varepsilon b = 0$. Write $b = \mu a$, so $\mu'a' = \beta\mu a = \mu'\alpha a \rightarrow a' = \alpha a$. Thus, $(a', b) = \{\alpha, \mu\}(a)$.

Claim: If $0 \rightarrow N \xrightarrow{\mu} P \xrightarrow{\varepsilon} A \rightarrow 0$ and $0 \rightarrow M \xrightarrow{\mu'} Q \xrightarrow{\varepsilon'} A \rightarrow 0$ are exact and P, Q are projective, then $P \oplus M \cong Q \oplus N$.

Proof: Since ε' is surjective, pick $\varphi' : P \rightarrow Q$ such that $\varepsilon'\varphi' = \varepsilon$. I claim that $\text{im } \varphi'\mu \leq \text{im } \mu'$. Indeed, $\text{im } \mu' = \ker \varepsilon'$ and $\varepsilon'\varphi'\mu = \varepsilon\mu = 0$. Thus, pick $\psi : N \rightarrow M$ such that $\varphi'\mu = \mu'\psi$. We have the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & N & \xrightarrow{\mu} & P & \xrightarrow{\varepsilon} & A \longrightarrow 0 \\ & & \downarrow \psi & & \downarrow \varphi' & & \parallel \\ 0 & \longrightarrow & M & \xrightarrow{\mu'} & Q & \xrightarrow{\varepsilon'} & A \longrightarrow 0 \end{array}$$

By the previous lemma, we have the exact sequence $0 \rightarrow N \rightarrow P \oplus M \rightarrow Q \rightarrow 0$. Since Q is projective, the sequence splits, so $P \oplus M \cong Q \oplus N$, as desired.

4 Free Modules over a PID

Claim: Every submodule of a free module over a PID is free.

Proof: Let $P = \bigoplus_{j \in J} \Lambda_j$ be free and J have a well-order. For $j \in J$, define $P_j = \bigoplus_{k \leq j} \Lambda_k$. Suppose $R \leq P$ is a submodule. Consider $R_j = P_j \cap R$. Denote by $\pi_j : P \rightarrow \Lambda$ the projection onto the j th coordinate. Let $I_j = \pi_j R_j \leq \Lambda$ as an ideal. Since Λ is a PID, let $I_j = (\alpha_j)$ for some $\alpha_j \in \Lambda$. Let $J' = \{j \in J : \alpha_j \neq 0\}$. For each $j \in J'$, pick $f_j \in R_j$ such that $\pi_j f_j = \alpha_j$. I claim that $\{f_j : j \in J'\}$ is a basis of R .

I claim that R is generated by the f_j . Let $x \in R$ and write $x = \sum_{i \in [n]} \lambda_{j_i} e_{j_i}$. Since $[n]$ is finite, we can pick $j = \max_{i \in [n]} j_i$, so $x \in R_j$. Pick $\lambda \in \Lambda$ such that $\pi_j(x) = \lambda \pi_j(f_j)$, so $\pi_j(x - \lambda f_j) = 0$. Thus, $x - \lambda f_j \in R_{j'}$ for some $j' < j$. If $x' = x - \lambda f_j = 0$, we're done. Else, repeat the process to x' . Suppose FTSOC we have an infinite descending chain $j > j' > j'' > \dots$. Since J is well-ordered, this is absurd.

Now suppose $\sum_{i \in [n]} \lambda_i f_{j_i} = 0$. Taking π_n of both sides, we have $\lambda_n \alpha_{j_n} = 0$. Since $\alpha_{j_n} \neq 0$, we have $\lambda_n = 0$. By induction, $\lambda_i = 0$ for all i .

Claim: Let Λ be a ring such that every left ideal is projective. Then every submodule of a free Λ -module is isomorphic to a sum of modules, each of which is isomorphic to an ideal of Λ .

Proof: Let $P = \bigoplus_{\beta < \alpha} \Lambda_\beta$. Suppose $\gamma \leq \alpha$. Define $P_\gamma = \bigoplus_{\alpha < \gamma} \Lambda_\alpha$. I claim that $P_\gamma = \bigcup_{\gamma' < \gamma} P_{\gamma'}$ iff γ is a limit ordinal. Suppose γ is a limit ordinal. Let $x \in P_\gamma$. Write $x = \sum_{i \in [n]} \lambda_i e_{\alpha_i}$. Let $\alpha = \max_{i \in [n]} \alpha_i$, so $x \in P_{\alpha+1}$. Since $\alpha < \gamma$ and γ is a limit, we have $\gamma' = \alpha + 1 < \gamma$, so $x \in \bigcup_{\gamma' < \gamma} P_{\gamma'}$. The other direction is clear. Now suppose γ is not a limit, so take γ_1 to be the predecessor of γ . Then $\bigcup_{\gamma' < \gamma} P_{\gamma'} \leq P_{\gamma_1}$. However, $e_{\gamma_1} \notin P_{\gamma_1}$.

Claim: Let Λ be a ring such that every left ideal is projective. Let P be a free module over Λ . Then P is a direct sum of submodules isomorphic to ideals of Λ .

Proof: Let $P = \bigoplus_{\alpha < \gamma_f} \Lambda_\alpha$, $P_\gamma = \bigoplus_{\alpha < \gamma} \Lambda_\alpha$, and $R \leq P$. Define $R_\gamma = P_\gamma \cap R$. I claim there exists an assignment γ to ideals I_γ of Λ such that $R_\gamma = \bigoplus_{\alpha < \gamma} I_\alpha$. The result is true for $\gamma = 0$ since $R_0 = 0$.

Suppose the result for γ . We show the result for $\gamma + 1$ if $\gamma + 1 \leq \gamma_f$. Let $\pi_\gamma : P \rightarrow \Lambda$ denote the projection onto the γ th coordinate. The sequence $0 \rightarrow (\ker \pi_\gamma) \cap R_{\gamma+1} \rightarrow R_{\gamma+1} \rightarrow \pi_\gamma R_{\gamma+1} \rightarrow 0$ is exact. $\pi_\gamma R_{\gamma+1}$ corresponds to an ideal $I_{\gamma+1}$ of Λ , so $\pi_\gamma R_{\gamma+1}$ is projective. Thus, the sequence splits and we have $R_{\gamma+1} \cong ((\ker \pi_\gamma) \cap R_{\gamma+1}) \oplus I_{\gamma+1}$. Note that $(\ker \pi_\gamma) \cap R_{\gamma+1} = R_\gamma$. By hypothesis, we have $R_{\gamma+1} \cong \bigoplus_{\alpha < \gamma+1} I_\alpha$, as desired.

Let $\gamma > 0$ be a limit ordinal. Since $P_\gamma = \bigcup_{\gamma' < \gamma} P_{\gamma'}$, we have $R_\gamma = \bigcup_{\gamma' < \gamma} R_{\gamma'}$. Then

$$R_\gamma = \bigcup_{\gamma' < \gamma} R_{\gamma'} = \bigcup_{\gamma' < \gamma} \bigoplus_{\alpha < \gamma'} I_\alpha = \bigoplus_{\alpha < \gamma} I_\alpha.$$

Claim: Let γ_f be a limit ordinal, $\{N_\gamma\}_{\gamma < \gamma_f}$ a collection of modules and $M_\gamma = \bigoplus_{\alpha < \gamma} N_\alpha$ for all $\gamma < \gamma_f$ and $M = M_{\gamma_f}$. Then $M = \bigcup_{\gamma < \gamma_f} M_\gamma$.

Proof: The proof is a more general version of the previous statement. We have $M_\gamma \leq M_{\gamma_f}$, so $\bigcup_{\gamma < \gamma_f} M_\gamma \leq M_{\gamma_f}$. Suppose $x \in M_{\gamma_f}$, so $x = \sum_{i \in [n]} \lambda_i x_{\alpha_i}$ for $x_{\alpha_i} \in N_{\alpha_i}$. Let $\alpha = \max_{i \in [n]} \alpha_i$, so $x \in M_{\alpha+1}$ (where $\alpha + 1 < \gamma_f$), so $x \in \bigcup_{\gamma < \gamma_f} M_\gamma$.

Claim: Let A, B, C be finitely generated modules over the PID Λ . If $A \oplus C \cong B \oplus C$, then $A \cong B$.

Proof: Write $A = R^r \oplus \bigoplus_{i \in [n]} R[x]/\langle a_i \rangle$, where $a_i \mid a_{i+1}$. Write similar equations for B, C . Comparing ranks, we have $r_A + r_C = r_B + r_C$, so $r_A = r_B$. Comparing torsions, we have $\bigoplus_{i \in [n]} R[x]/\langle a_i \rangle \oplus \bigoplus_{i \in [l]} R[x]/\langle c_i \rangle \cong \bigoplus_{i \in [m]} R[x]/\langle b_i \rangle \oplus \bigoplus_{i \in [l]} R[x]/\langle c_i \rangle$. Reindexing so that we can take $a_i = 1$ and that the a_i, b_i, c_i are indexed by $[n]$, and aligning to the “right,” we have $a_i b_i = b_i c_i \rightarrow a_i = b_i$ up to a unit for all i . Hence, $A \cong B$.

Suppose the modules need not be finitely generated. Take $A = 0, B = \mathbb{Z}, C = \mathbb{Z}^{\oplus \mathbb{N}_0}$. Suppose Λ is not a PID. Take Λ to be the ring from before. Then $\Lambda \cong \Lambda \oplus \Lambda$, so we can take $A = 0, B = \Lambda, C = \Lambda$.

Submodules of projective modules need not be projective. Take $\Lambda = \mathbb{Z}_{p^2}$. We have the short exact sequence $0 \rightarrow \mathbb{Z}_p \rightarrow \mathbb{Z}_{p^2} \rightarrow \mathbb{Z}_p \rightarrow 0$. If $\mathbb{Z}_p$ were projective, then $\mathbb{Z}_{p^2} \cong \mathbb{Z}_p^2$ as modules over $\mathbb{Z}_{p^2}$, absurd (by order considerations). Note the order of an element $x \in M$ for a module M over $\mathbb{Z}_n$ is the minimal $k \in \mathbb{Z}^+$ such that $kx = 0$.

5 Injective Modules over a PID

Claim: Any abelian group A can be embedded in a divisible abelian group D .

Proof: Consider $a \in A$. Define a map $\alpha : (a) \rightarrow \mathbb{Q}/\mathbb{Z}$ as follows. If $a = 0$, take $\alpha a = 0$. If $a \neq 0$ has order ∞ , take αa to be any nonzero element. If $a \neq 0$ has finite order $|a|$, take $\alpha a = 1/|a|$. We have the embedding $(a) \rightarrow A$. Since $\mathbb{Q}/\mathbb{Z}$ is injective, extend to $\psi_a : A \rightarrow \mathbb{Q}/\mathbb{Z}$. By extensionality, let $\psi : A \rightarrow \prod_{a \in A} \mathbb{Q}/\mathbb{Z}$.

I claim that ψ is injective. Suppose $\psi(a) = 0$. Then $\psi_a(a) = 0$. Assume $a \neq 0$. If a has infinite order, then $\psi_a(a) = \alpha a \neq 0$, absurd. If a has finite order, then $\psi_a(a) = 1/|a| \neq 0$ as well.

Claim: The Λ module I injective iff for every $J \trianglelefteq \Lambda$ (where $\trianglelefteq$ denotes left ideal, and Λ need not be commutative) and $\alpha : J \rightarrow I$, there exists a map $\beta : \Lambda \rightarrow I$ such that the following diagram commutes:

$$\begin{array}{ccc} J & \longrightarrow & \Lambda \\ \downarrow \alpha & \nearrow \beta & \\ I & & \end{array}$$

Proof: One direction is clear, so consider the other. Treat $A \leq B$. Let $\mathcal{S} = \{(A', \alpha') : A \leq A' \leq B, \alpha' : A' \rightarrow I\}$ have the partial order $(A', \alpha') \leq (A'', \alpha'')$ if $A' \leq A''$ and $\alpha''|_{A'} = \alpha'$. By Zorn's lemma let $(\bar{A}, \bar{\alpha})$ be maximal with respect to this partial order.

Suppose FTSOC $\bar{A} < B$. Pick $b \in B - \bar{A}$ and define $J = \{\lambda \in \Lambda : \lambda b \in \bar{A}\}$. I claim that $J \trianglelefteq \Lambda$. Suppose $\lambda_1, \lambda_2 \in J$. Then $(\lambda_1 - \lambda_2)b = \lambda_1 b - \lambda_2 b \in \bar{A}$, so $\lambda_1 - \lambda_2 \in J$. Suppose $r \in \Lambda$ is arbitrary and $\lambda \in J$. Then $r\lambda b = r(\lambda b) \in \bar{A}$ since $\bar{A} \leq B$. Thus, $J \trianglelefteq \Lambda$.

Define $\varphi : J \rightarrow I$ by $\varphi(\lambda) = \bar{\alpha}(\lambda b)$. By Extensionality, pick $\tilde{\varphi} : \Lambda \rightarrow I$ extending φ . Define $\tilde{\alpha} : \langle \bar{A}, b \rangle \rightarrow I$ by $\tilde{\alpha}(\bar{a} + \lambda b) = \bar{\alpha}(\bar{a}) + \tilde{\varphi}(\lambda)$, where $\lambda \in \Lambda$

is arbitrary. I claim that $\tilde{\alpha}$ is well-defined. Suppose $\overline{a_1} + \lambda_1 b = \overline{a_2} + \lambda_2 b$, so $\overline{a_1} - \overline{a_2} = (\lambda_2 - \lambda_1)b$, so $\lambda_2 - \lambda_1 \in J$. Applying $\overline{\alpha}$ to both sides, we have $\overline{\alpha}(\overline{a_1} - \overline{a_2}) = \varphi(\lambda_2 - \lambda_1)$. The RHS rewrites as $\tilde{\varphi}(\lambda_2 - \lambda_1) = \tilde{\varphi}(\lambda_2) - \tilde{\varphi}(\lambda_1)$, so $\overline{\alpha}(\overline{a_1}) + \tilde{\varphi}(\lambda_1) = \overline{\alpha}(\overline{a_2}) + \tilde{\varphi}(\lambda_2)$. One can check that $\tilde{\alpha}$ is indeed a homomorphism extending $\overline{\alpha}$, contradicting maximality.

Let $0 \rightarrow R \rightarrow F \rightarrow A \rightarrow 0$ be a short exact sequence of abelian groups with F free. By embedding F into a direct sum of copies of $\mathbb{Q}$, show how to embed A into a divisible group.

6 Cofree Modules

Let A be a right Λ module and G an abelian group. Define the action of Λ on $\text{Hom}_{\mathbb{Z}}(A, G)$ by $(\lambda\varphi)(a) = \varphi(a\lambda)$. We check

$$\begin{aligned} (\lambda(\varphi_1 + \varphi_2))(a) &= (\varphi_1 + \varphi_2)(a\lambda) = \varphi_1(a\lambda) + \varphi_2(a\lambda) = (\lambda\varphi_1)(a) + (\lambda\varphi_2)(a) \\ ((\lambda_1 + \lambda_2)\varphi)(a) &= \varphi(a(\lambda_1 + \lambda_2)) = \varphi(a\lambda_1 + a\lambda_2) = \varphi(a\lambda_1) + \varphi(a\lambda_2) \\ ((\lambda_1\lambda_2)\varphi)(a) &= \varphi(a(\lambda_1\lambda_2)) = \varphi((a\lambda_1)\lambda_2) = (\lambda_2\varphi)(a\lambda_1) = (\lambda_1(\lambda_2\varphi))(a) \\ (1\varphi)(a) &= \varphi(a1) = \varphi(a) \end{aligned}$$

Let A be a left Λ module and G an abelian group. Give Λ a right Λ module structure. Then we have the group isomorphism

$$\eta = \eta_A : \text{Hom}_{\Lambda}(A, \text{Hom}_{\mathbb{Z}}(\Lambda, G)) \rightarrow \text{Hom}_{\mathbb{Z}}(A, G)$$

Let $\varphi : A \rightarrow \text{Hom}_{\mathbb{Z}}(\Lambda, G)$. I claim η is given by $\eta(\varphi)(a) = \varphi(a)(1)$. We have $\eta(\varphi_1 + \varphi_2)(a) = (\varphi_1 + \varphi_2)(a)(1) = (\varphi_1(a) + \varphi_2(a))(1) = \varphi_1(a)(1) + \varphi_2(a)(1)$. Suppose $\eta(\varphi) = 0$, so $\eta(\varphi)(a) = \varphi(a)(1) = 0$ for all $a \in A$, so $\varphi(a) = 0$ for all a , so $\varphi = 0$. Now let $\psi : A \rightarrow G$ be a group homomorphism. Pick $\varphi : A \rightarrow \text{Hom}_{\mathbb{Z}}(\Lambda, G)$ by $\varphi(a)(\lambda) = \psi(\lambda a)$. We have $\varphi(a)(\lambda_1 + \lambda_2) = \varphi((\lambda_1 + \lambda_2)a) = \varphi(\lambda_1 a) + \varphi(\lambda_2 a) = \varphi(a)(\lambda_1) + \varphi(a)(\lambda_2)$, so $\varphi(a) \in \text{Hom}_{\mathbb{Z}}(\Lambda, G)$. I claim that φ is a left Λ module homomorphism. Indeed,

$$\begin{aligned} \varphi(a + a')(\lambda) &= \psi(\lambda(a + a')) = \psi(\lambda a + \lambda a') = \psi(\lambda a) + \psi(\lambda a') = \varphi(a)(\lambda) + \varphi(a')(\lambda) \\ \varphi(\lambda_0 a)(\lambda) &= \psi(\lambda(\lambda_0 a)) = \psi((\lambda\lambda_0)a) = \varphi(a)(\lambda\lambda_0) = (\lambda_0\varphi(a))(\lambda), \end{aligned}$$

so η is surjective.

Further, we claim that the following diagram is commutative for $\alpha : A \rightarrow B$:

$$\begin{array}{ccc} \text{Hom}_{\Lambda}(B, \text{Hom}_{\mathbb{Z}}(\Lambda, G)) & \xrightarrow{\eta_B} & \text{Hom}_{\mathbb{Z}}(B, G) \\ \downarrow \alpha^* & & \downarrow \alpha^* \\ \text{Hom}_{\Lambda}(A, \text{Hom}_{\mathbb{Z}}(\Lambda, G)) & \xrightarrow{\eta_A} & \text{Hom}_{\mathbb{Z}}(A, G) \end{array}$$

Take $\varphi : B \rightarrow \text{Hom}_{\mathbb{Z}}(\Lambda, G)$. We compute $\alpha^* \eta_B(\varphi)(a) = \eta_B(\varphi)(\alpha a) = \varphi(\alpha a)(1)$ and $(\eta_A \alpha^*(\varphi))(a) = (\alpha^* \varphi)(a)(1) = (\varphi \alpha(a))(1)$.

Claim: Let G be a divisible abelian group. Then $\bar{\Lambda} = \text{Hom}_{\mathbb{Z}}(\Lambda, G)$ is an injective Λ -module.

We first prove a lemma. Suppose we have the following commutative diagram:

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ & \searrow \alpha & \swarrow \beta \\ & \bar{\Lambda} & \end{array}$$

Then $\eta_A(\alpha) = \eta_B(\beta)\varphi$. Indeed, let $a \in A$. Then

$$\begin{aligned} \eta_A(\alpha)(a) &= \alpha(a)(1) = (\beta\varphi)(a)(1) = \beta(\varphi a)(1) \\ &= \eta_B(\beta)\varphi(a), \end{aligned}$$

as desired.

Suppose $\alpha : A \rightarrow \bar{A}$ is a module homomorphism and $\varphi = \mu$ is injective. We wish to show the existence of β . Denote by $\alpha' = \eta_A(\alpha)$. Consider the diagram

$$\begin{array}{ccc} A & \xrightarrow{\mu} & B \\ & \searrow \alpha' & \swarrow \beta' \\ & G & \end{array}$$

for μ injective. Since G is divisible over the PID $\mathbb{Z}$, G is injective. Thus, we can pick a group homomorphism β' extending α' . Pick $\beta' = \eta_B(\beta)$. Then we have $\eta_A(\alpha) = \eta_B(\beta)\mu = \eta_A(\beta\varphi)$. Since η_A is injective, we have $\alpha = \beta\varphi$.

Claim: Every Λ module can be embedded in a cofree, thus injective, module.

Proof: (This only shows that we can embed into an injective module). Let the Λ -module A be embedded as an abelian group into the divisible abelian group D . We have the embeddings

$$A \rightarrow \text{Hom}_{\Lambda}(\Lambda, A) \rightarrow \text{Hom}_{\mathbb{Z}}(\Lambda, D),$$

so A is embedded in the injective module $\text{Hom}_{\mathbb{Z}}(\Lambda, D)$.

For a Λ -module I , the following are equivalent:

- (i). I is injective.
- (ii). For every short exact sequence, $0 \rightarrow A \xrightarrow{\mu} B \xrightarrow{\varepsilon} C \rightarrow 0$, $0 \rightarrow \text{Hom}_{\Lambda}(C, I) \xrightarrow{\varepsilon^*} \text{Hom}_{\Lambda}(B, I) \xrightarrow{\mu^*} \text{Hom}_{\Lambda}(A, I) \rightarrow 0$ is exact.

- (iii). For every injective $\mu : I \rightarrow B$, there exists $\beta : B \rightarrow I$ such that $\beta\mu = 1_I$.
- (iv). I is a direct summand (factor) in every module for which it is a submodule.
- (v). I is a direct summand of a cofree module.

(i) implies (ii). We wish to show $\mu^* : \text{Hom}_\Lambda(B, I) \rightarrow \text{Hom}_\Lambda(A, I)$ is surjective. Let $\alpha : A \rightarrow I$. Since $\mu : A \rightarrow B$ is injective and I is injective, pick $\beta : B \rightarrow I$ extending α : $\alpha = \beta\mu = \mu^*(\beta)$.

(ii) implies (iii). The sequence $0 \rightarrow I \xrightarrow{\mu} B \xrightarrow{p} \text{coker } \mu \rightarrow 0$ is exact. We have that $0 \rightarrow \text{Hom}_\Lambda(\text{coker } \mu, I) \xrightarrow{p^*} \text{Hom}_\Lambda(B, I) \xrightarrow{\mu^*} \text{Hom}_\Lambda(I, I) \rightarrow 0$ is exact. Take $1_I \in \text{Hom}_\Lambda(I, I)$. Since μ^* is surjective, pick $\beta \in \text{Hom}_\Lambda(B, I)$ such that $\mu^*(\beta) = 1_I$, ie $\beta\mu = 1_I$.

(iii) implies (iv). Take $\mu : I \rightarrow A$ to be inclusion. Pick $\alpha : A \rightarrow I$ such that $\alpha\mu = 1_I$. Thus, we have the split exact sequence $0 \rightarrow I \xrightarrow{\mu} A \rightarrow \text{coker } \mu \rightarrow 0$, so $A \cong I \oplus \text{coker } \mu$.

(iv) implies (v). Embed $I \rightarrow J$, where J is cofree, so J is in particular injective. Write $J = I \oplus K$ by (iv).

(v) implies (i). Since J is injective and finite direct products are the same as direct sums, we have that I is injective.

For a left Λ -module A , give $A^* = \text{Hom}_\Lambda(A, \mathbb{Q}/\mathbb{Z})$ a right Λ -module structure. Show that A can be embedded in A^{**} .

Proof: Note that A^{**} has left-module structure. Define $i : A \rightarrow A^{**}$ by $i(a)(\varphi) = \varphi(a)$ for $\varphi \in A^* = \text{Hom}_\mathbb{Z}(A, \mathbb{Q}/\mathbb{Z})$. I claim that i is well-defined. Indeed, $i(a)(\varphi_1 + \varphi_2) = (\varphi_1 + \varphi_2)(a) = \varphi_1(a) + \varphi_2(a) = i(a)(\varphi_1) + i(a)(\varphi_2)$. We check that i is a homomorphism:

$$\begin{aligned} i(a_1 + a_2)(\varphi) &= \varphi(a_1 + a_2) = \varphi(a_1) + \varphi(a_2) = i(a_1)(\varphi) + i(a_2)(\varphi) \\ i(\lambda a)(\varphi) &= \varphi(\lambda a) = (\varphi\lambda)(a) = i(a)(\varphi\lambda) = (\lambda i(a))(\varphi). \end{aligned}$$

Suppose $i(a) = 0$. Then $\varphi(a) = 0$ for all $\varphi \in \text{Hom}_\mathbb{Z}(A, \mathbb{Q}/\mathbb{Z})$. Suppose FTSOC $a \neq 0$. If a has infinite order, take $(a) \rightarrow \mathbb{Q}/\mathbb{Z}$ by mapping a to a nonzero element. Else, take $(a) \rightarrow \mathbb{Q}/\mathbb{Z}$ by mapping a to $1/|a|$. Since $\mathbb{Q}/\mathbb{Z}$ is divisible over $\mathbb{Z}$, $\mathbb{Q}/\mathbb{Z}$ is injective. Extend to $\varphi : A \rightarrow \mathbb{Q}/\mathbb{Z}$. Then $\varphi(a) \neq 0$, absurd.

Show that the above embedding allows us to embed A into a cofree module. Let $0 \rightarrow N \xrightarrow{j} F \xrightarrow{p} A^* \rightarrow 0$ be a free presentation of A^* . The sequence $0 \rightarrow A^{**} = \text{Hom}_\Lambda(A^*, \mathbb{Q}/\mathbb{Z}) \xrightarrow{p^*} \text{Hom}_\Lambda(F, \mathbb{Q}/\mathbb{Z}) \xrightarrow{j^*} \text{Hom}_\Lambda(N, \mathbb{Q}/\mathbb{Z}) \rightarrow 0$ is exact.

Write $F = \Lambda^{\oplus J}$. Then we have the embedding

$$\begin{aligned} A \rightarrow A^{**} &\xrightarrow{p^*} \text{Hom}_\Lambda(F, \mathbb{Q}/\mathbb{Z}) = \prod_J \text{Hom}_\Lambda(\Lambda, \mathbb{Q}/\mathbb{Z}) \\ &\rightarrow \prod_J \text{Hom}_\mathbb{Z}(\Lambda, \mathbb{Q}/\mathbb{Z}) = (\Lambda^*)^{\Pi^J} \end{aligned}$$

7 Essential Extensions

A monomorphism $\mu : A \rightarrow B$ is called essential if for $0 < H \leq B$ that $\mu A \cap H \neq 0$. Ex: $\mathbb{Q}$ is an essential extension of $\mathbb{Z}$. Suppose $0 < H \leq \mathbb{Q}$. Pick nonzero $q = m/n \in H$, so $nq \in H \cap \mathbb{Z}$ is nonzero.

Essential extensions are transitive. Suppose $\mu_1 : A_1 \rightarrow A_2, \mu_2 : A_2 \rightarrow A_3$ are monomorphisms. Then $\mu_2\mu_1 : A_1 \rightarrow A_3$ is a monomorphism. Let $0 < H \leq A_3$.

I claim that equivalently, $\mu^{-1}H \neq 0$ if $H \neq 0$. Suppose $\mu : A \rightarrow B$ is essential and $H \neq 0$, so $\mu A \cap H \neq 0$. Pick $h \in \mu A \cap H$, so $h = \mu a$ and $h \in H$ is nonzero. Then $a \neq 0$, so $a = \mu^{-1}h \in \mu^{-1}H$, so $\mu^{-1}H \neq 0$. Suppose $\mu^{-1}H \neq 0$ for all $H \neq 0$. Pick nonzero $a \in \mu^{-1}H$, so $\mu a \in H$ is nonzero since μ is injective.

Now $\mu_2^{-1}H \neq 0$, so $\mu_1^{-1}\mu_2^{-1}H = (\mu_2\mu_1)^{-1}H \neq 0$. Thus, $\mu_2\mu_1 : A_1 \rightarrow A_3$ is injective.

Claim: B is an essential extension of A iff for every $0 \neq b \in B$, there exists $\lambda \in \Lambda$ such that $0 \neq \lambda b \in A$.

Suppose B is essential. Let $H = \langle b \rangle \neq 0$ since $b \neq 0$, so $H \cap A \neq 0$, ie there exists $0 \neq \lambda b \in A$. Conversely, suppose for all $0 \neq b \in B$, there exists λ such that $0 \neq \lambda b \in A$. Let $0 \neq H \leq B$. Pick $0 \neq b \in H$. Pick $0 \neq \lambda b \in A$, so $\lambda b \in H \cap A$.

Let $A \leq M$ and Φ denote the set of essential extensions of A contained in M . Let $\mathcal{C}$ denote a chain of essential extensions. Let $\bar{A} = \bigcup \mathcal{C} \leq M$. Suppose $0 \neq x \in \bar{A}$, so $x \in A_\alpha$ for some α . There exists λ such that $0 \neq \lambda x \in A_\alpha \leq \bar{A}$, so $\bar{A}$ is essential. Thus, Φ is inductive.

Claim: Let $A \leq I$ and E a maximal essential extension of A in I . Then E is injective.

First, we show that E has no nontrivial essential extensions. Suppose $\mu : E \rightarrow X$ is a monomorphism. Let $j : E \rightarrow I$ be inclusion. Since X is injective, pick $\zeta : X \rightarrow I$ such that $\zeta\mu = j$. Suppose $\text{FTSOC } \ker \zeta \neq 0$. Since X/E is essential, pick $0 \neq x \in X$ such that $\zeta x = 0$ and $x = \mu e$ for some $e \in E$. Then $\zeta x = \zeta\mu e = je = e$. Since $x \neq 0$, we have $e \neq 0$, absurd. Thus, ζ is a monomorphism. $\text{im } \zeta/X/E/A \rightarrow \text{im } \zeta/A$ is essential, so $\text{im } \zeta = E$.

Let $\mathcal{S} = \{H \leq I : H \cap E = 0\}$. Note $\mathcal{S} \neq \emptyset$ since $0 \in \mathcal{S}$. $\mathcal{S}$ has the usual partial order. Suppose $\emptyset < \mathcal{C} \leq \mathcal{S}$ is a chain. Then $\bigcup \mathcal{C} \leq I$ and $\bigcup \mathcal{C} \cap E = \bigcup \emptyset = \emptyset$. Thus, $\mathcal{S}$ is inductive, so let $\bar{H}$ be a maximal element. I claim that $I \cong E \oplus \bar{H}$. Consider the projection $\pi : I \rightarrow I/\bar{H}$. Let $\sigma = \pi|_E : E \rightarrow I/\bar{H}$ denote the restriction. Note that σ is a monomorphism. I claim that σ is an essential extension. Suppose $0 < \tilde{H} < I/\bar{H}$. Let $H = \pi^{-1}\tilde{H}$, so we have $\bar{H} < H \leq I$. By maximality of $\bar{H}$, we have $H \cap E \neq 0$, so $\pi H \cap \pi E \neq 0 \rightarrow \tilde{H} \cap \sigma E \neq 0$.

By the previous result, σ is an isomorphism. Consider the exact sequence $0 \rightarrow \bar{H} \rightarrow I \xrightarrow{\pi} I/\bar{H} \rightarrow 0$. Thus, we have the exact sequence $0 \rightarrow \bar{H} \rightarrow I \xrightarrow{\sigma^{-1}\pi} E \rightarrow 0$. I claim that $\sigma^{-1}\pi j = 1_E$, or $\pi j = \sigma$, which is indeed true. Thus, the sequence splits and $I \cong \bar{H} \oplus E$. Since I is injective, so is E .

Let $A \leq I_1, I_2$. Pick E_1, E_2 maximal essential extensions of A contained in I_1, I_2 . Then $E_1 \cong E_2$.

Proof: Consider the diagram

$$\begin{array}{ccc} A & \longrightarrow & E_1 \\ \downarrow & \searrow \phi & \\ E_2 & & \end{array}$$

Since the map $A \rightarrow E_2$ is injective and $A \rightarrow E_1$ is essential, we have $\phi : E_1 \rightarrow E_2$ is a monomorphism. E_2/E_1 is essential, so ϕ is an isomorphism. Since every module can be embedded in an injective module, the injective envelope exists.

Suppose $A_2/A_1/A_0$ such that A_2/A_0 is essential. Then I claim $A_2/A_1, A_1/A_0$ are essential. Suppose $0 < H < A_2$. Then $H \cap A_0 \neq 0 \rightarrow H \cap A_1 \neq 0$. Now suppose $0 < H \leq A_1$. Then $0 < H \leq A_2$, so $H \cap A_0 \neq 0$.

I claim that the injective envelope of $\mathbb{Z}$ is $\mathbb{Q}$.

Proof: Consider the inclusion $j : \mathbb{Z} \rightarrow \mathbb{Q}$. Let $\mathbb{Z} \leq E \leq \mathbb{Q}$. Since E is injective over $\mathbb{Z}$, E is divisible. For $z \neq 0$ in $\mathbb{Z}$, pick $a_z \in E$ such that $za_z = 1$. In $\mathbb{Q}$, we have $a_z = 1/z$, so by closure under addition, we have $E = \mathbb{Q}$. Note $\mathbb{Q}/\mathbb{Z}$ is indeed essential.

I claim that the injective envelope of $\mathbb{Z}_p$ is $T_p(\mathbb{Q}/\mathbb{Z})$.

Proof: First, we show that $T_p(\mathbb{Q}/\mathbb{Z})$ is divisible. Let $\bar{x} \in T_p(\mathbb{Q}/\mathbb{Z})$ and $z \neq 0$ in $\mathbb{Z}$, so $x = m/p^n$. Write $z = p^k r$ for $p \nmid r$. Pick $s \in \mathbb{Z}$ such that $rs \equiv 1 \pmod{p^n}$. Then

$$z \cdot \frac{ms}{p^{n+k}} = \frac{mrs}{p^n} \equiv \frac{m}{p^n} = x \pmod{1},$$

so $T_p(\mathbb{Q}/\mathbb{Z})$ is injective. Suppose $\mathbb{Z}_p \leq E \leq T_p(\mathbb{Q}/\mathbb{Z})$. For $n \in \mathbb{Z}^+$, pick e_n such that $p^n e_n = \overline{1/p}$. Let $e_n = \frac{m}{p^k}$ with $p \nmid m$ (note $m \neq 0$, so this is well-defined). We have $\frac{mp^n}{p^k} - \frac{1}{p} \in \mathbb{Z}$. Note $k > n$. Suppose FTSOC $k > n + 1$. Write

$$\frac{mp^n}{p^k} - \frac{1}{p} = \frac{m}{p^{k-n}} - \frac{1}{p} = \frac{m}{p^{k-n}} - \frac{p^{k-n-1}}{p^{k-n}} = \frac{m - p^{k-n-1}}{p^{k-n}},$$

so $p \mid m - p^{k-n-1} \rightarrow p \mid m$, absurd. Thus, $k = n + 1$. It follows $(m-1)/p \in \mathbb{Z}$, so $m \equiv 1 \pmod{p}$. Assume $e_n = \overline{1/p^{n+1}}$ works. e_0 is given. Write $m = pm_1 + 1$. Then $\frac{m}{p^{n+1}} = \frac{m_1}{p^n} + \frac{1}{p^{n+1}}$, so $e_n = \overline{1/p^{n+1}} \in E$, so $E = T_p(\mathbb{Q}/\mathbb{Z})$.

Finally, we wish to show $T_p(\mathbb{Q}/\mathbb{Z})$ is an essential extension of $\mathbb{Z}_p = \overline{(1/p)}$. Suppose $x = m/p^n$ with $p \nmid m$ and $n > 0$ (for $n = 0$, we have $\bar{x} = 0$). Then

$$p^{n-1}\bar{x} = p^{n-1}\overline{m/p^n} = m\overline{(1/p)} \in \mathbb{Z}_p$$

since $p \nmid m$, we have $m\overline{(1/p)} \neq 0$, as desired.

Compute the injective envelope E of $\mathbb{Z}_n$. Suppose n has the distinct prime factors $p_1, p_2, \dots, p_r$. Then $E = \bigoplus_{i \in [r]} T_{p_i}(\mathbb{Q}/\mathbb{Z})$. The $T_{p_i}(\mathbb{Q}/\mathbb{Z})$ have zero intersection. Suppose $\overline{m/p^n} \in T_p(\mathbb{Q}/\mathbb{Z})$ and $\overline{k/q^l} \in T_q(\mathbb{Q}/\mathbb{Z})$ with $p \nmid m$, $q \nmid k$, $n, l \geq 1$ such that $\overline{m/p^n} = \overline{k/q^l}$. Then

$$\frac{m}{p^n} - \frac{k}{q^l} = \frac{mq^l - kp^n}{p^n q^l} \in \mathbb{Z}$$

so $p \mid mq^l \rightarrow p \mid m$, absurd. We only care about the presence of the primes since $p^{n-1}(\overline{1/p^n}) = \overline{1/p}$.

Let B_i/A_i be essential. Then $B_1 \oplus B_2$ is an essential extension of $A_1 \oplus A_2$.

Proof: Let $b = b_1 + b_2 \in B_1 \oplus B_2$. If either of b_1, b_2 is zero, we're done. Assume $b_1, b_2 \neq 0$. Pick $\lambda_1 \in \Lambda$ such that $0 \neq \lambda_1 b_1 \in A_1$. Then $\lambda_1 b = \lambda_1 b_1 + \lambda_1 b_2$. If $\lambda_1 b_2 = 0$, we're done. Else, pick λ_2 such that $0 \neq \lambda_2(\lambda_1 b_2)$. Thus, $\lambda_2 \lambda_1 b = \lambda_2 \lambda_1 b_1 + \lambda_2(\lambda_1 b_2) \neq 0$ is in $A_1 \oplus A_2$. We use that submodules are closed under multiplication by Λ .

More generally, let B_j/A_j be essential. Then $B = \bigoplus_{j \in J} B_j$ is an essential extension of $A = \bigoplus_{j \in J} A_j$. Let $0 \neq b = (b_j) \in B$. Let $J' = \{j \in J : b_j \neq 0\}$, so J' is finite. Enumerate $J' = \{j_1, j_2, \dots, j_n\}$, where $j_i < j_{i+1}$. I claim the result by induction on n . The result is true for $n = 1$. Pick λ_1 such that $0 \neq \lambda_1 b_{j_1} \in A_1$. Then the index j_1 of $b' = \lambda_1 b$ is nonzero and in A_1 . If the other components are zero, we're done. Else, pick the next index j_k with $b_{j_k} \neq 0$ and apply the result to the components of b from index j_k onward.

Claim: The dual to the injective envelope does not exist. Suppose $\varepsilon : P \rightarrow \mathbb{Z}_5$ is

surjective, where P is projective. Since P is projective over the PID $\mathbb{Z}$, P is free. Consider $\eta : \mathbb{Z} \rightarrow \mathbb{Z}_5$ given by composing $\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$ with the natural projection $\mathbb{Z} \rightarrow \mathbb{Z}_5$. Pick epimorphism $\varepsilon' : \mathbb{Z} \rightarrow P$ such that $\varepsilon\varepsilon' = \eta$. Thus, $P \cong \mathbb{Z}$ and $\varepsilon' = \pm 1$. It follows $\varepsilon = \pm\eta$, absurd.